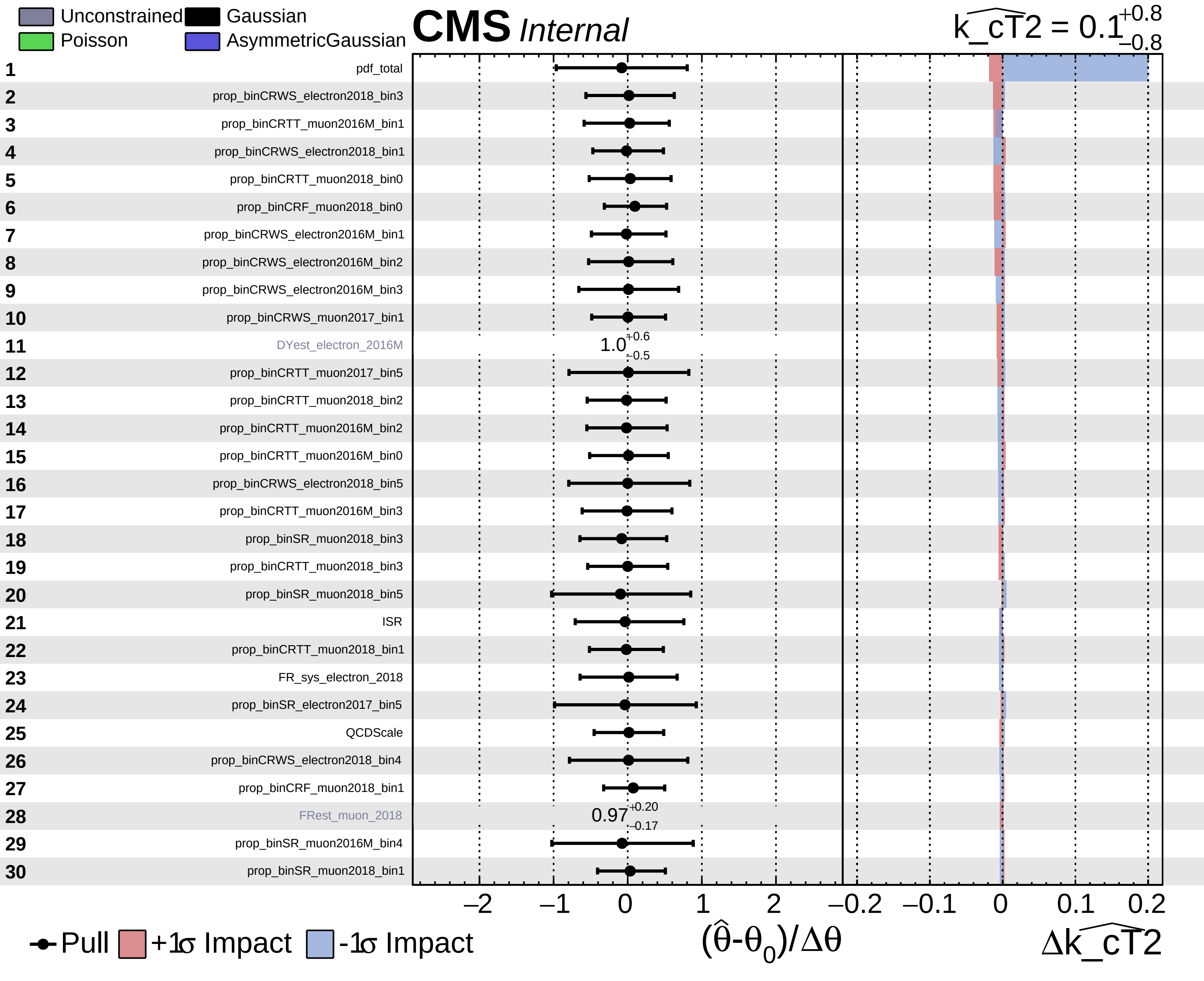
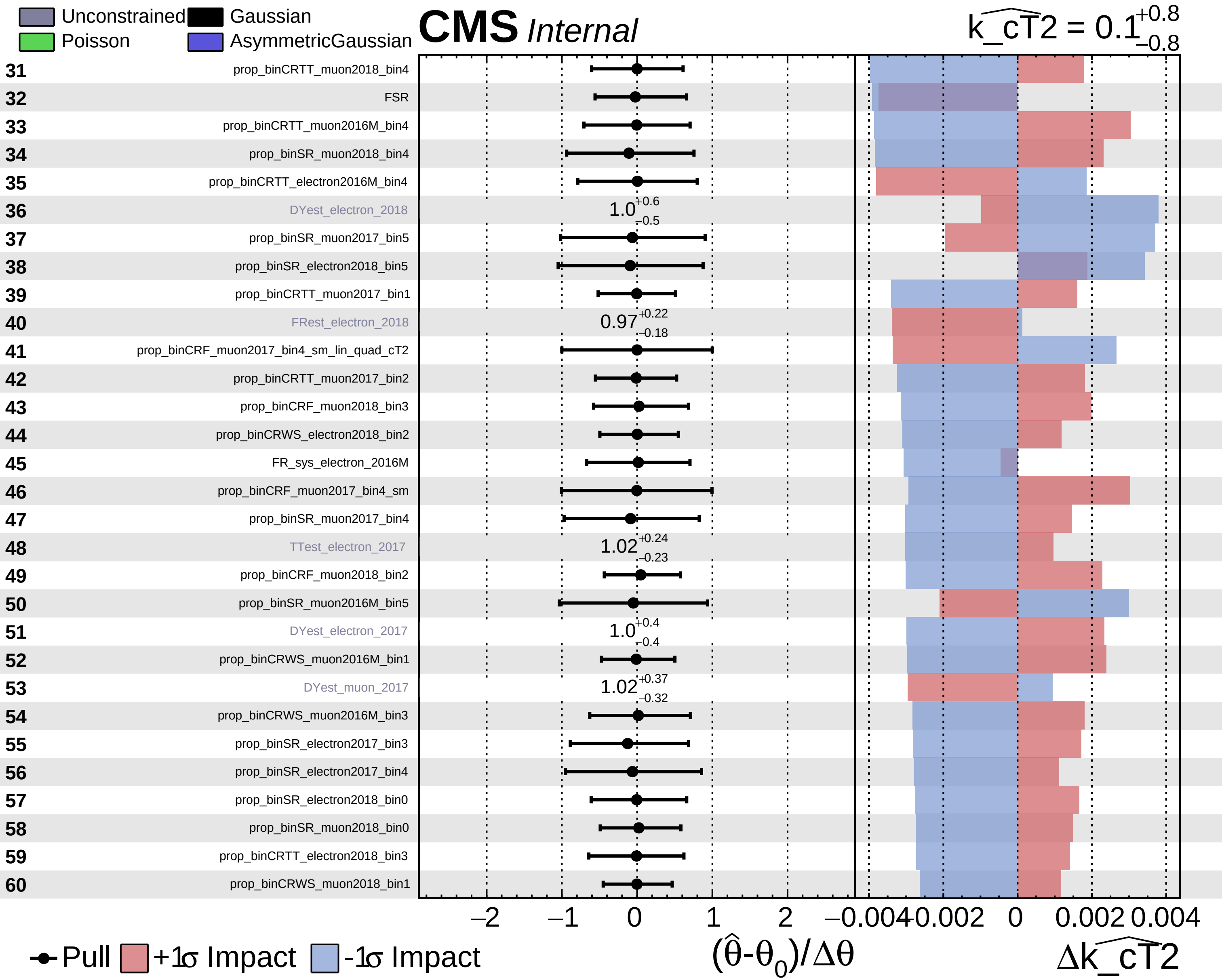
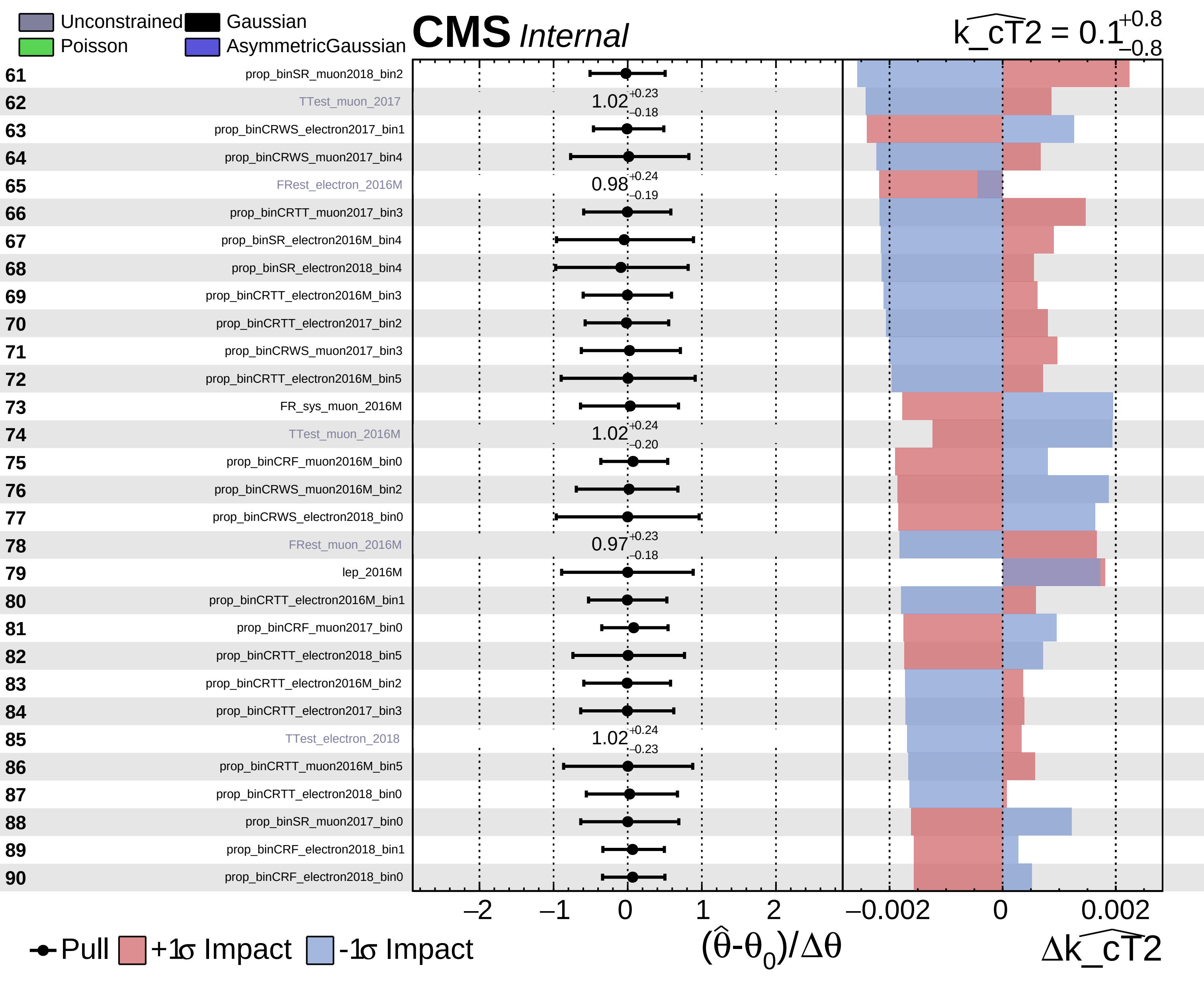
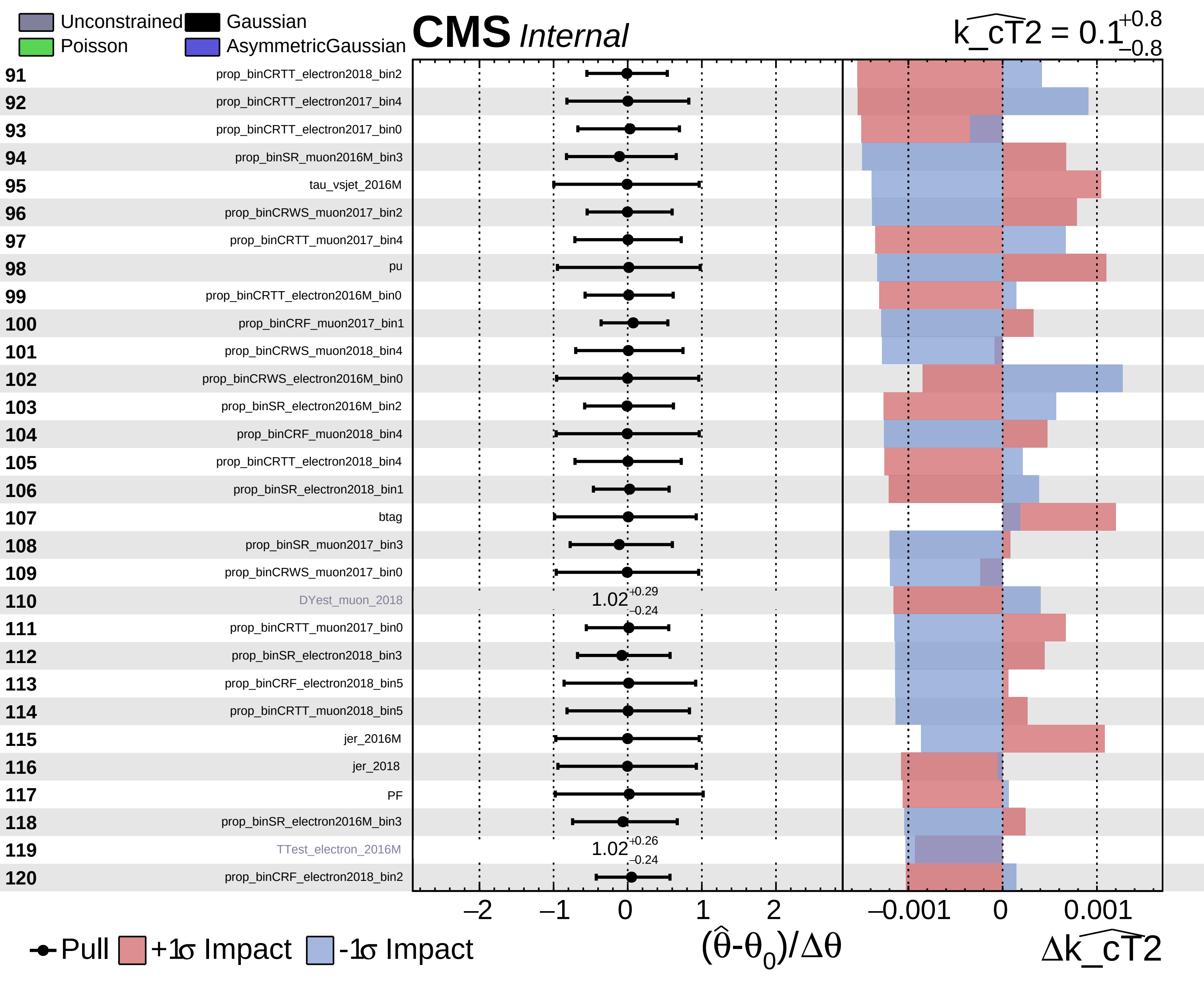


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.1 +0.8 −0.8
1 pdf_total
2 prop_binCRWS_electron2018_bin3
3 prop_binCRTT_muon2016M_bin1
4 prop_binCRWS_electron2018_bin1
5 prop_binCRTT_muon2018_bin0
6 prop_binCRF_muon2018_bin0
7 prop_binCRWS_electron2016M_bin1
8 prop_binCRWS_electron2016M_bin2
9 prop_binCRWS_electron2016M_bin3
10 prop_binCRWS_muon2017_bin1
11 DYest_electron_2016M
1.0 +0.6 −0.5
12 prop_binCRTT_muon2017_bin5
13 prop_binCRTT_muon2018_bin2
14 prop_binCRTT_muon2016M_bin2
15 prop_binCRTT_muon2016M_bin0
16 prop_binCRWS_electron2018_bin5
17 prop_binCRTT_muon2016M_bin3
18 prop_binSR_muon2018_bin3
19 prop_binCRTT_muon2018_bin3
20 prop_binSR_muon2018_bin5
21 ISR
22 prop_binCRTT_muon2018_bin1
23 FR_sys_electron_2018
24 prop_binSR_electron2017_bin5
25 QCDScale
26 prop_binCRWS_electron2018_bin4
27 prop_binCRF_muon2018_bin1
28 FRest_muon_2018
0.97 +0.20 −0.17
29 prop_binSR_muon2016M_bin4
30 prop_binSR_muon2018_bin1
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

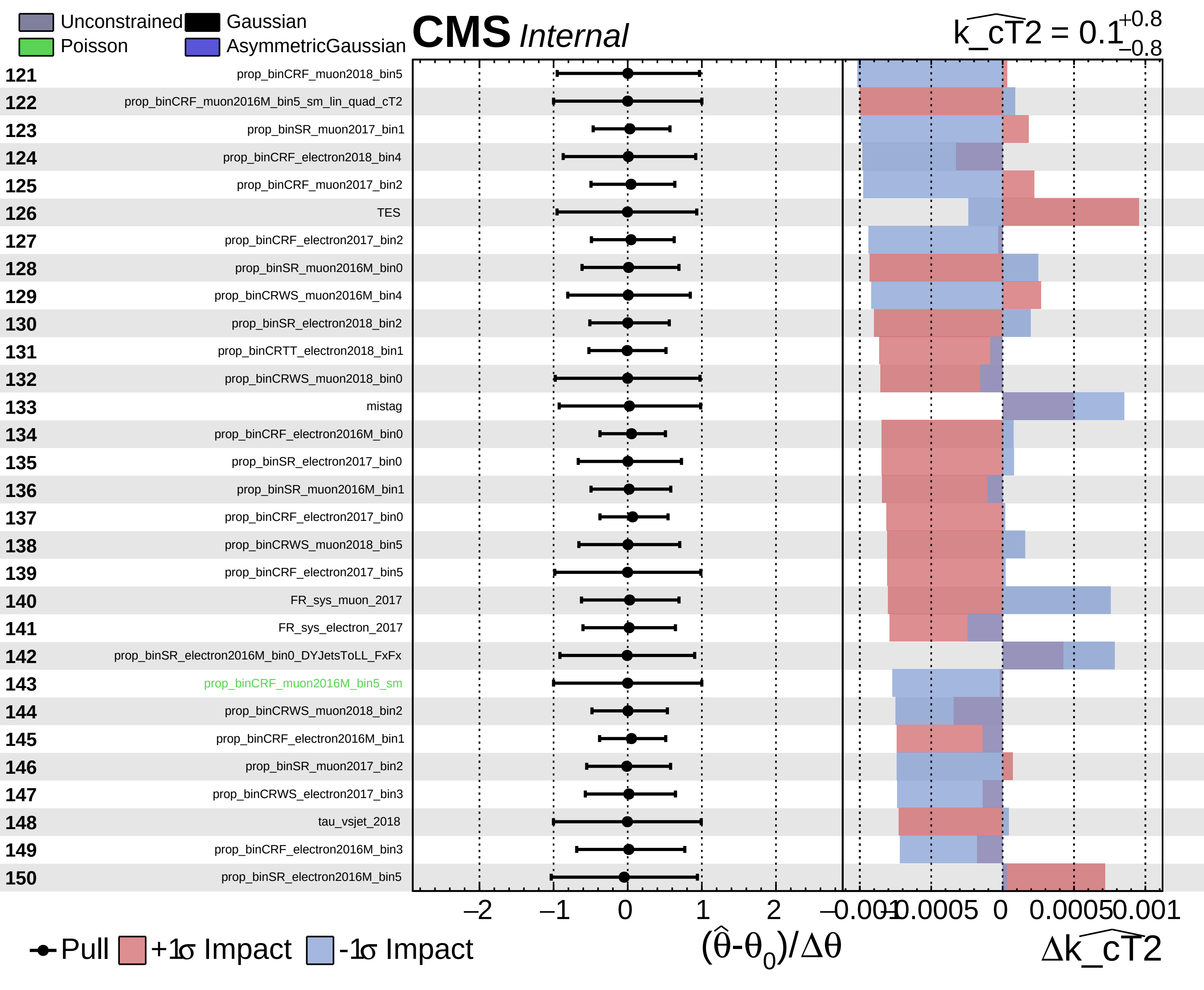

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.1 +0.8 −0.8
31 prop_binCRTT_muon2018_bin4
32 FSR
33 prop_binCRTT_muon2016M_bin4
34 prop_binSR_muon2018_bin4
35 prop_binCRTT_electron2016M_bin4
36 DYest_electron_2018 1.0 +0.6 −0.5
37 prop_binSR_muon2017_bin5
38 prop_binSR_electron2018_bin5
39 prop_binCRTT_muon2017_bin1
40 FRest_electron_2018 0.97 +0.22 −0.18
41 prop_binCRF_muon2017_bin4_sm_lin_quad_cT2
42 prop_binCRTT_muon2017_bin2
43 prop_binCRF_muon2018_bin3
44 prop_binCRWS_electron2018_bin2
45 FR_sys_electron_2016M
46 prop_binCRF_muon2017_bin4_sm
47 prop_binSR_muon2017_bin4
48 TTest_electron_2017 1.02 +0.24 −0.23
49 prop_binCRF_muon2018_bin2
50 prop_binSR_muon2016M_bin5
51 DYest_electron_2017 1.0 +0.4 −0.4
52 prop_binCRWS_muon2016M_bin1
53 DYest_muon_2017 1.02 +0.37 −0.32
54 prop_binCRWS_muon2016M_bin3
55 prop_binSR_electron2017_bin3
56 prop_binSR_electron2017_bin4
57 prop_binSR_electron2018_bin0
58 prop_binSR_muon2018_bin0
59 prop_binCRTT_electron2018_bin3
60 prop_binCRWS_muon2018_bin1
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



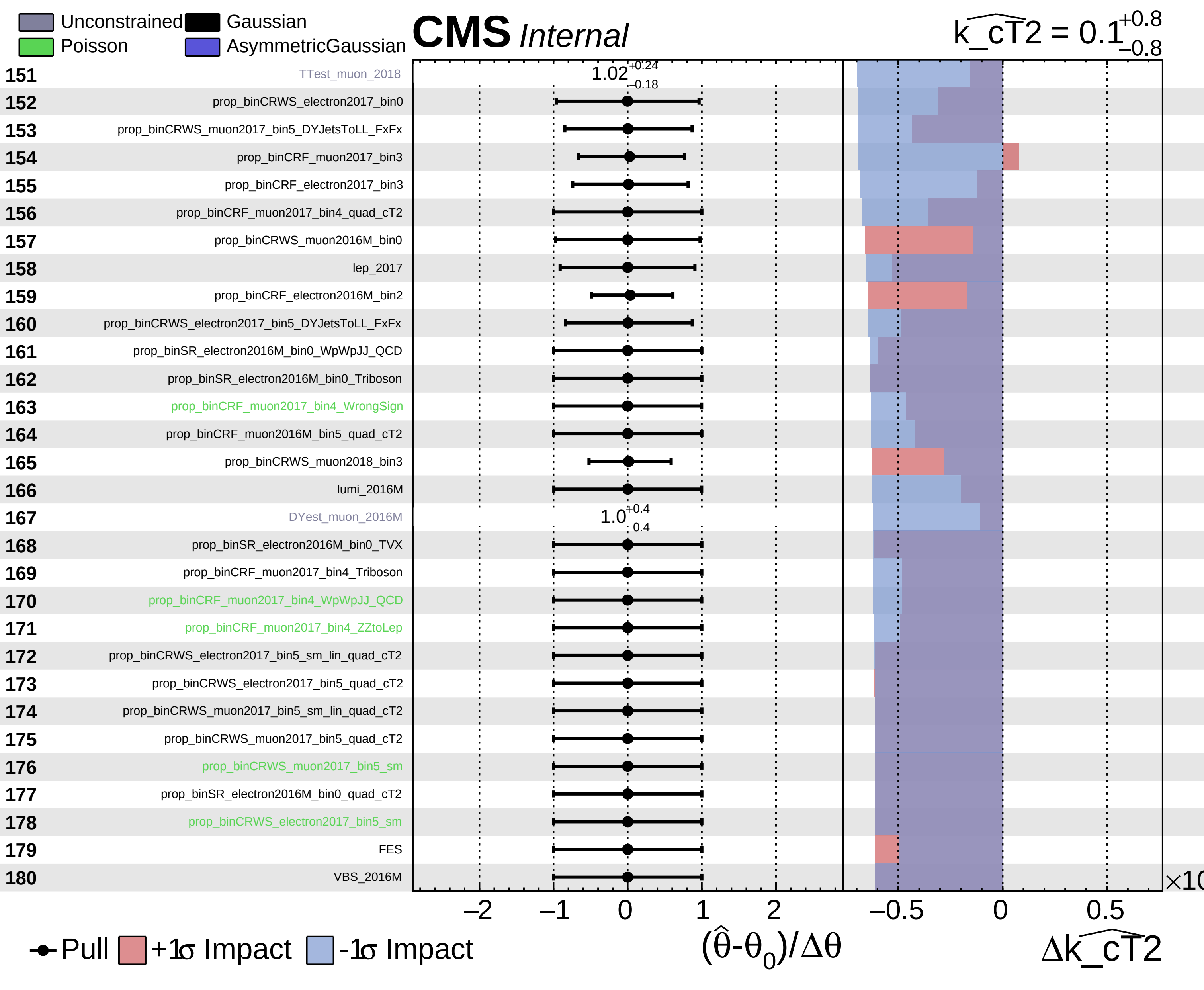
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.1 +0.8 −0.8
61 prop_binSR_muon2018_bin2
62 TTest_muon_2017 1.02 +0.23 −0.18
63 prop_binCRWS_electron2017_bin1
64 prop_binCRWS_muon2017_bin4
65 FRest_electron_2016M 0.98 +0.24 −0.19
66 prop_binCRTT_muon2017_bin3
67 prop_binSR_electron2016M_bin4
68 prop_binSR_electron2018_bin4
69 prop_binCRTT_electron2016M_bin3
70 prop_binCRTT_electron2017_bin2
71 prop_binCRWS_muon2017_bin3
72 prop_binCRTT_electron2016M_bin5
73 FR_sys_muon_2016M
74 TTest_muon_2016M 1.02 +0.24 −0.20
75 prop_binCRF_muon2016M_bin0
76 prop_binCRWS_muon2016M_bin2
77 prop_binCRWS_electron2018_bin0
78 FRest_muon_2016M 0.97 +0.23 −0.18
79 lep_2016M
80 prop_binCRTT_electron2016M_bin1
81 prop_binCRF_muon2017_bin0
82 prop_binCRTT_electron2018_bin5
83 prop_binCRTT_electron2016M_bin2
84 prop_binCRTT_electron2017_bin3
85 TTest_electron_2018 1.02 +0.24 −0.23
86 prop_binCRTT_muon2016M_bin5
87 prop_binCRTT_electron2018_bin0
88 prop_binSR_muon2017_bin0
89 prop_binCRF_electron2018_bin1
90 prop_binCRF_electron2018_bin0
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2



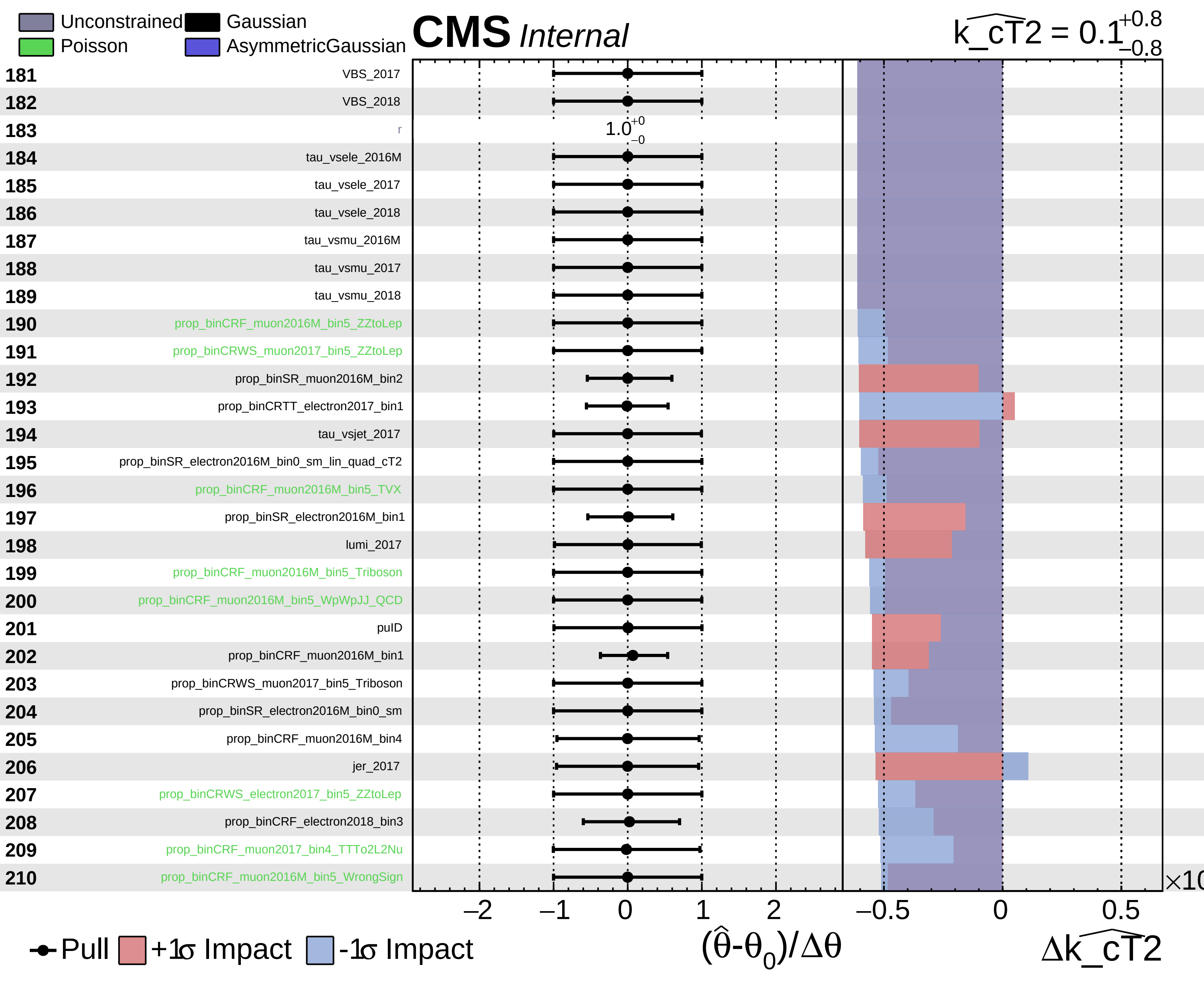
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.1 +0.8 −0.8
91 prop_binCRTT_electron2018_bin2
92 prop_binCRTT_electron2017_bin4
93 prop_binCRTT_electron2017_bin0
94 prop_binSR_muon2016M_bin3
95 tau_vsjet_2016M
96 prop_binCRWS_muon2017_bin2
97 prop_binCRTT_muon2017_bin4
98 pu
99 prop_binCRTT_electron2016M_bin0
100 prop_binCRF_muon2017_bin1
101 prop_binCRWS_muon2018_bin4
102 prop_binCRWS_electron2016M_bin0
103 prop_binSR_electron2016M_bin2
104 prop_binCRF_muon2018_bin4
105 prop_binCRTT_electron2018_bin4
106 prop_binSR_electron2018_bin1
107 btag
108 prop_binSR_muon2017_bin3
109 prop_binCRWS_muon2017_bin0
110 DYest_muon_2018
1.02 +0.29 −0.24
111 prop_binCRTT_muon2017_bin0
112 prop_binSR_electron2018_bin3
113 prop_binCRF_electron2018_bin5
114 prop_binCRTT_muon2018_bin5
115 jer_2016M
116 jer_2018
117 PF
118 prop_binSR_electron2016M_bin3
119 TTest_electron_2016M
1.02 +0.26 −0.24
120 prop_binCRF_electron2018_bin2
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



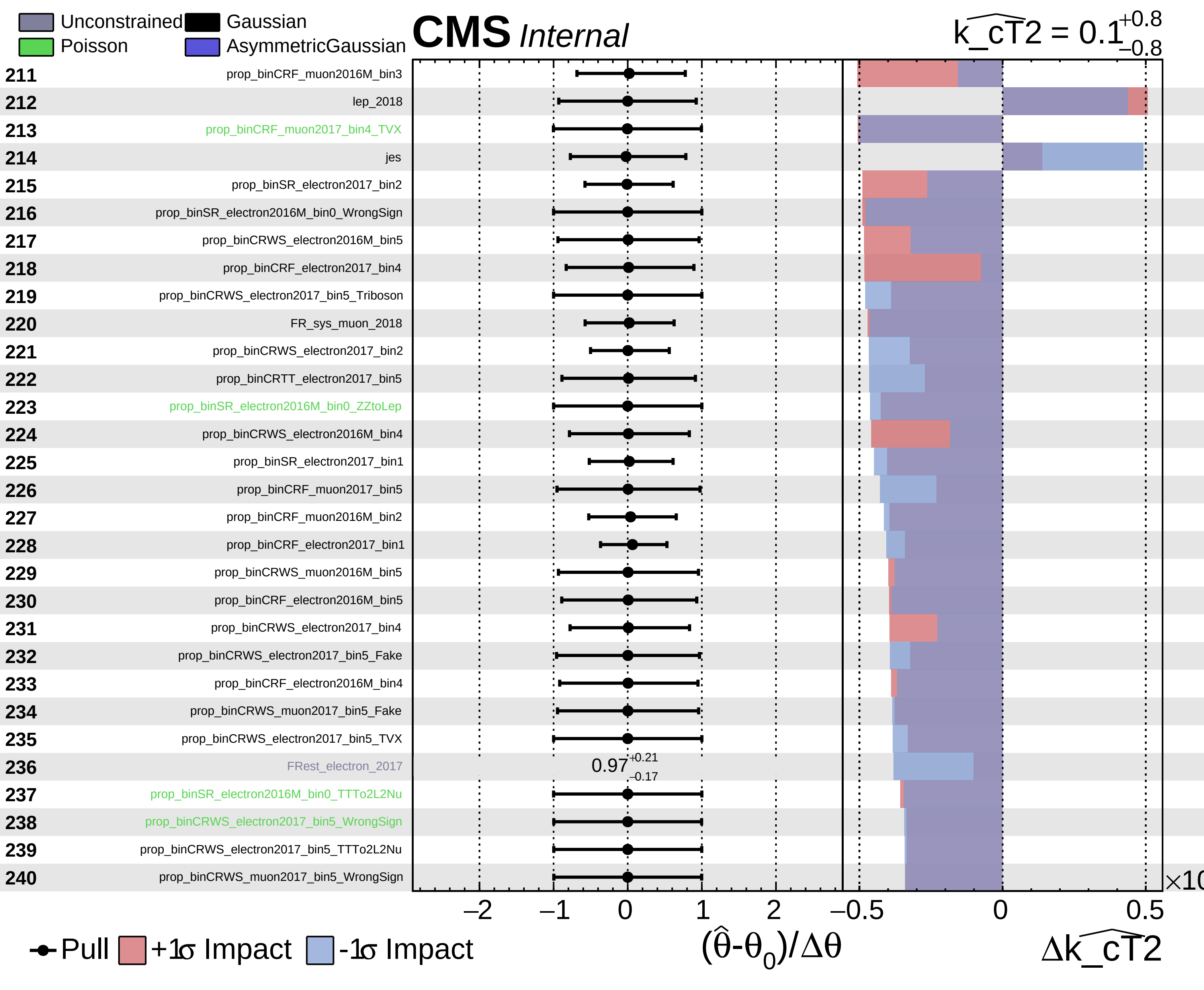
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.1 +0.8 −0.8
121 prop_binCRF_muon2018_bin5
122 prop_binCRF_muon2016M_bin5_sm_lin_quad_cT2
123 prop_binSR_muon2017_bin1
124 prop_binCRF_electron2018_bin4
125 prop_binCRF_muon2017_bin2
126 TES
127 prop_binCRF_electron2017_bin2
128 prop_binSR_muon2016M_bin0
129 prop_binCRWS_muon2016M_bin4
130 prop_binSR_electron2018_bin2
131 prop_binCRTT_electron2018_bin1
132 prop_binCRWS_muon2018_bin0
133 mistag
134 prop_binCRF_electron2016M_bin0
135 prop_binSR_electron2017_bin0
136 prop_binSR_muon2016M_bin1
137 prop_binCRF_electron2017_bin0
138 prop_binCRWS_muon2018_bin5
139 prop_binCRF_electron2017_bin5
140 FR_sys_muon_2017
141 FR_sys_electron_2017
142 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
143 prop_binCRF_muon2016M_bin5_sm
144 prop_binCRWS_muon2018_bin2
145 prop_binCRF_electron2016M_bin1
146 prop_binSR_muon2017_bin2
147 prop_binCRWS_electron2017_bin3
148 tau_vsjet_2018
149 prop_binCRF_electron2016M_bin3
150 prop_binSR_electron2016M_bin5
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2



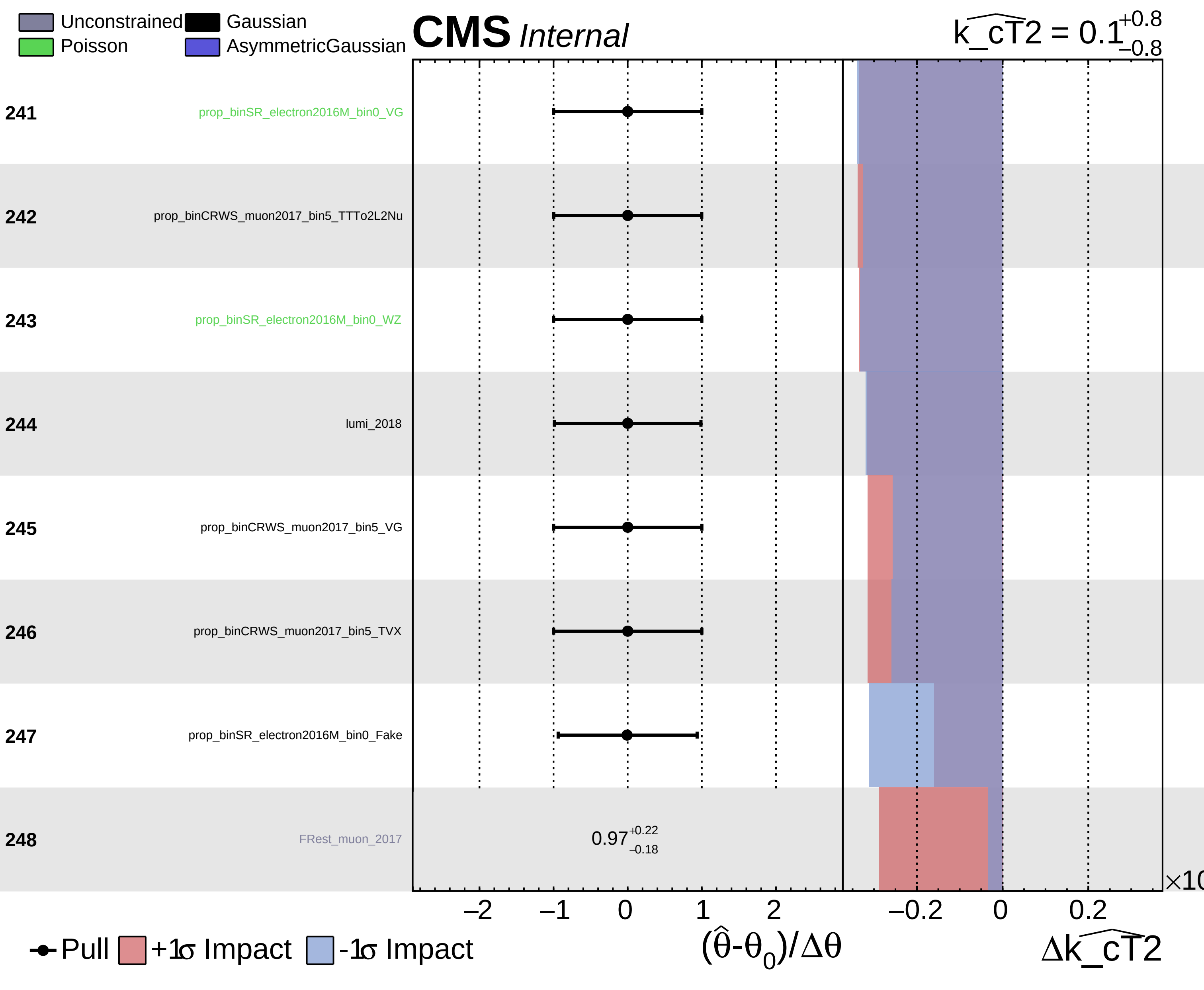
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.1 +0.8 −0.8
151 TTest_muon_2018
1.02 +0.24 −0.18
152 prop_binCRWS_electron2017_bin0
153 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
154 prop_binCRF_muon2017_bin3
155 prop_binCRF_electron2017_bin3
156 prop_binCRF_muon2017_bin4_quad_cT2
157 prop_binCRWS_muon2016M_bin0
158 lep_2017
159 prop_binCRF_electron2016M_bin2
160 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
161 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
162 prop_binSR_electron2016M_bin0_Triboson
163 prop_binCRF_muon2017_bin4_WrongSign
164 prop_binCRF_muon2016M_bin5_quad_cT2
165 prop_binCRWS_muon2018_bin3
166 lumi_2016M
167 DYest_muon_2016M
1.0 +0.4 −0.4
168 prop_binSR_electron2016M_bin0_TVX
169 prop_binCRF_muon2017_bin4_Triboson
170 prop_binCRF_muon2017_bin4_WpWpJJ_QCD
171 prop_binCRF_muon2017_bin4_ZZtoLep
172 prop_binCRWS_electron2017_bin5_sm_lin_quad_cT2
173 prop_binCRWS_electron2017_bin5_quad_cT2
174 prop_binCRWS_muon2017_bin5_sm_lin_quad_cT2
175 prop_binCRWS_muon2017_bin5_quad_cT2
176 prop_binCRWS_muon2017_bin5_sm
177 prop_binSR_electron2016M_bin0_quad_cT2
178 prop_binCRWS_electron2017_bin5_sm
179 FES
180 VBS_2016M
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.1^{+0.8}_{-0.8}$
181 VBS_2017
182 VBS_2018
183 r
1.0^{+0}_{-0}
184 tau_vsele_2016M
185 tau_vsele_2017
186 tau_vsele_2018
187 tau_vsmu_2016M
188 tau_vsmu_2017
189 tau_vsmu_2018
190 prop_binCRF_muon2016M_bin5_ZZtoLep
191 prop_binCRWS_muon2017_bin5_ZZtoLep
192 prop_binSR_muon2016M_bin2
193 prop_binCRTT_electron2017_bin1
194 tau_vsjet_2017
195 prop_binSR_electron2016M_bin0_sm_lin_quad_cT2
196 prop_binCRF_muon2016M_bin5_TVX
197 prop_binSR_electron2016M_bin1
198 lumi_2017
199 prop_binCRF_muon2016M_bin5_Triboson
200 prop_binCRF_muon2016M_bin5_WpWpJJ_QCD
201 puID
202 prop_binCRF_muon2016M_bin1
203 prop_binCRWS_muon2017_bin5_Triboson
204 prop_binSR_electron2016M_bin0_sm
205 prop_binCRF_muon2016M_bin4
206 jer_2017
207 prop_binCRWS_electron2017_bin5_ZZtoLep
208 prop_binCRF_electron2018_bin3
209 prop_binCRF_muon2017_bin4_TTTo2L2Nu
210 prop_binCRF_muon2016M_bin5_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.1 +0.8 −0.8
211 prop_binCRF_muon2016M_bin3
212 lep_2018
213 prop_binCRF_muon2017_bin4_TVX
214 jes
215 prop_binSR_electron2017_bin2
216 prop_binSR_electron2016M_bin0_WrongSign
217 prop_binCRWS_electron2016M_bin5
218 prop_binCRF_electron2017_bin4
219 prop_binCRWS_electron2017_bin5_Triboson
220 FR_sys_muon_2018
221 prop_binCRWS_electron2017_bin2
222 prop_binCRTT_electron2017_bin5
223 prop_binSR_electron2016M_bin0_ZZtoLep
224 prop_binCRWS_electron2016M_bin4
225 prop_binSR_electron2017_bin1
226 prop_binCRF_muon2017_bin5
227 prop_binCRF_muon2016M_bin2
228 prop_binCRF_electron2017_bin1
229 prop_binCRWS_muon2016M_bin5
230 prop_binCRF_electron2016M_bin5
231 prop_binCRWS_electron2017_bin4
232 prop_binCRWS_electron2017_bin5_Fake
233 prop_binCRF_electron2016M_bin4
234 prop_binCRWS_muon2017_bin5_Fake
235 prop_binCRWS_electron2017_bin5_TVX
236 FRest_electron_2017
0.97 +0.21 −0.17
237 prop_binSR_electron2016M_bin0_TTTo2L2Nu
238 prop_binCRWS_electron2017_bin5_WrongSign
239 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
240 prop_binCRWS_muon2017_bin5_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.1 +0.8 −0.8
241
prop_binSR_electron2016M_bin0_VG
242
prop_binCRWS_muon2017_bin5_TTTo2L2Nu
243
prop_binSR_electron2016M_bin0_WZ
244
lumi_2018
245
prop_binCRWS_muon2017_bin5_VG
246
prop_binCRWS_muon2017_bin5_TVX
247
prop_binSR_electron2016M_bin0_Fake
248
FRest_muon_2017
0.97 +0.22 −0.18
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2